APPENDIX B

Electromagnetic Moments and Transitions

We can learn much about the structure of nuclei by studying their behavior in an external electromagnetic field. The interaction of the nucleus with such a field is very well known, and is therefore a very direct way of obtaining information about a nucleus. Many authors have derived the corresponding formulae and we therefore refer the reader to the literature for the details (for example, [BW 52, Da 65a, SF 74]).

B.1 The General Form of the Hamiltonian

The system of a nucleus moving in an external electromagnetic field characterized by its four-dimensional potential $A_\mu = (\Phi, \mathbf{A})$ and the corresponding electromagnetic fields $\mathbf{E}$ and $\mathbf{B}$ is described by the Hamiltonian

$$H = H_{\text{nucl}} + H_{\text{field}} + H_{\text{int}}. \tag{B.1}$$

It has three parts: H_{nucl} is the Hamiltonian of the nucleus, or at least some nuclear model Hamiltonian. We assume that we know its eigenfunctions $\Psi_i(1 \ldots A)$

$$H_{\text{nucl}} \Psi_i(1 \ldots A) = E_i \Psi_i(1 \ldots A). \tag{B.2}$$

H_{field} is given by [Ja 62]

$$H_{\text{field}} = \frac{1}{8\pi} \int \left(\mathbf{E}^2(\mathbf{r}, t) + \mathbf{B}^2(\mathbf{r}, t) \right) d^3r. \tag{B.3}$$

The interaction between the external field and the nucleus is

$$H_{\text{int}} = -\frac{1}{c}\int j_\mu A^\mu d^3r = \int \left(\rho(\mathbf{r},t)\Phi(\mathbf{r},t) - \frac{1}{c}\mathbf{j}(\mathbf{r},t)\cdot\mathbf{A}(\mathbf{r},t)\right) d^3r. \quad \text{(B.4)}$$

$\rho(\mathbf{r},t)$ is the nuclear *charge density*:

$$\rho(\mathbf{r},t) = \sum_{i=1}^{A} e\left(\tfrac{1}{2} - t_3^{(i)}\right)\cdot\delta(\mathbf{r}-\mathbf{r}_i(t)). \quad \text{(B.5)}$$

Here we have considered the nucleons as point particles. Actually, they have a finite extension which is given by a form factor. This could easily be introduced here [BM 69, App. 3C], but for reasons of simplicity we will disregard it in the following.

The *current density* $\mathbf{j}(\mathbf{r},t)$ in the nucleus is connected to the density $\boldsymbol{\mu}(\mathbf{r},t)$ of the magnetic moment [Ja 62, Eq. 5.81] by

$$\mathbf{j}(\mathbf{r},t) = c\nabla\times\boldsymbol{\mu}(\mathbf{r},t). \quad \text{(B.6)}$$

It has two parts:

(i) An *orbit part*, which comes from the moving charges of the protons

$$j^o(\mathbf{r},t) = \sum_{i=1}^{A} e\left(\tfrac{1}{2} - t_3^{(i)}\right)\frac{1}{2}(\mathbf{v}_i\delta(\mathbf{r}-\mathbf{r}_i(t)) + \text{h.c.}). \quad \text{(B.7)}$$

$\mathbf{v}_i$ is given by

$$\mathbf{v}_i = \frac{i}{\hbar}[H, \mathbf{r}_i]. \quad \text{(B.8)}$$

For velocity independent potentials, we get $\mathbf{v}_i = (1/m)\mathbf{p}_i$. In the following we will only consider this case. Again, (B.7) is an approximation that does not take into account the exchange current of pions; for a discussion of this point, see [SF 74].

(ii) *A spin part*, which has its origins in the spin distribution of the nucleus and is given by (B.6) and

$$\boldsymbol{\mu}^s(\mathbf{r},t) = \sum_{i=1}^{A}\delta(\mathbf{r}-\mathbf{r}_i(t))\frac{e\hbar}{2mc}\left\{\left(\tfrac{1}{2} - t_3^{(i)}\right)g_p + \left(\tfrac{1}{2} + t_3^{(i)}\right)g_n\right\}\mathbf{s}_i, \quad \text{(B.9)}$$

where g_p and g_n are the g-factors for free protons and neutrons ($g_p = 5.586$ and $g_n = -3.826$). Corrections for the exchange currents of pions are sometimes taken into account by using effective g-values. In the following, we use the nuclear magneton $\mu_N = e\hbar/2mc$.

B.2 Static Multipole Moments

The interaction with a static electromagnetic field causes a change of the nuclear energy in a state i, which is given in first-order perturbation theory by

$$\Delta E_i = \langle\Psi_i|H_{\text{int}}|\Psi_i\rangle. \quad \text{(B.10)}$$

We introduce the density of the magnetic flux $\mathbf{B}=\nabla\times\mathbf{A}$ and we get from (B.4) and (B.6)

$$H_{\text{int}}=\int\rho(\mathbf{r})\Phi(\mathbf{r})\,d^3r-\int\boldsymbol{\mu}(\mathbf{r})\cdot\mathbf{B}(\mathbf{r})\,d^3r. \tag{B.11}$$

Since the sources of the external field are far from the nucleus, the quantities $\Phi(r)$ and $\mathbf{B}(r)$ fulfill the homogeneous Maxwell equations in the nuclear region.

$$\begin{aligned}\Delta\Phi(\mathbf{r})&=0,\\ \nabla\times\mathbf{B}(\mathbf{r})&=0,\\ \nabla\cdot\mathbf{B}(\mathbf{r})&=0.\end{aligned} \tag{B.12}$$

This means that $\mathbf{B}$ can be written as a gradient,

$$\mathbf{B}(\mathbf{r})=-\nabla\Xi(\mathbf{r}), \tag{B.13}$$

with

$$\Delta\Xi(\mathbf{r})=0. \tag{B.14}$$

In spherical coordinates, the most general solution of the Laplace equation (B.12) and (B.14) which is not singular at the origin is given by

$$\Phi(\mathbf{r})=\sum_{\lambda\mu}a_{\lambda\mu}r^{\lambda}Y_{\lambda\mu}(\theta,\varphi), \tag{B.15}$$

$$\Xi(\mathbf{r})=\sum_{\lambda\mu}b_{\lambda\mu}r^{\lambda}Y_{\lambda\mu}(\theta,\varphi), \tag{B.16}$$

where $Y_{\lambda\mu}$ are spherical harmonics [Ed 57].

We find

$$H_{\text{int}}=\sum_{\lambda\mu}a_{\lambda\mu}\hat{Q}_{\lambda\mu}+b_{\lambda\mu}\hat{M}_{\lambda\mu} \tag{B.17}$$

with the electric and magnetic multipole operators

$$\hat{Q}_{\lambda\mu}=\int\rho(\mathbf{r})r^{\lambda}Y_{\lambda\mu}(\theta,\varphi)\,d^3r, \tag{B.18}$$

$$\hat{M}_{\lambda\mu}=\int\boldsymbol{\mu}(\mathbf{r})\cdot\nabla\left(r^{\lambda}Y_{\lambda\mu}(\theta,\varphi)\right)d^3r. \tag{B.19}$$

Before inserting the expressions (B.6) and (B.9) for the density and the current, we have to rewrite the operator $\hat{M}_{\lambda\mu}$ a little.

We use the relation

$$(\nabla\times(\nabla\times\mathbf{r}))r^{\lambda}Y_{\lambda\mu}(\theta,\varphi)=(\lambda+1)\nabla\left(r^{\lambda}Y_{\lambda\mu}(\theta,\varphi)\right) \tag{B.20}$$

and obtain, after integrating (B.19) by parts [the surface terms vanish because $\boldsymbol{\mu}(\mathbf{r})$ is localized in the nucleus]:

$$\hat{M}_{\lambda\mu}=\frac{1}{\lambda+1}\int(\mathbf{r}\times[\nabla\times\boldsymbol{\mu}(\mathbf{r})])\cdot[\nabla r^{\lambda}Y_{\lambda\mu}]\,d^3r \tag{B.21}$$

$$=\frac{1}{c(\lambda+1)}\int(\mathbf{r}\times\mathbf{j}(\mathbf{r}))\cdot[\nabla r^{\lambda}Y_{\lambda\mu}]\,d^3r. \tag{B.22}$$

We now insert (B.5) into (B.18), (B.7) into (B.22), and (B.9) into (B.19), and find with $\mathbf{l}_i = (1/i)(\mathbf{r}_i \times \nabla_i)$,

$$\hat{Q}_{\lambda\mu} = e \sum_{i=1}^{A} \left(\tfrac{1}{2} - t_3^{(i)}\right) r_i^{\lambda} Y_{\lambda\mu}(\theta_i, \varphi_i), \tag{B.23}$$

$$\hat{M}_{\lambda\mu} = \mu_N \sum_{i=1}^{A} \left\{ g_s^{(i)} \mathbf{s}_i + \frac{2}{\lambda+1} \cdot g_l^{(i)} \mathbf{l}_i \right\} \cdot \left(\nabla r^{\lambda} Y_{\lambda\mu}(\theta, \varphi) \right)_{\mathbf{r}=\mathbf{r}_i}, \tag{B.24}$$

with

$$\left.\begin{aligned} g_s^{(i)} &= g_p \\ g_l^{(i)} &= 1 \end{aligned}\right\} \quad \text{for protons,} \qquad \left.\begin{aligned} g_s^{(i)} &= g_n \\ g_l^{(i)} &= 0 \end{aligned}\right\} \quad \text{for neutrons.}$$

In (B.24), the operator $\mathbf{l}_i$ acts only on the coordinate $\mathbf{r}_i$ in the wave function. It commutes with $\nabla r^{\lambda} Y_{\lambda\mu}$ because

$$\mathbf{l} \cdot \left(\nabla r^{\lambda} Y_{\lambda\mu} \right) \propto (\mathbf{r} \times \nabla) \cdot \nabla r^{\lambda} Y_{\lambda\mu} = \mathbf{r} \cdot (\nabla \times \nabla) r^{\lambda} Y_{\lambda\mu} = 0. \tag{B.25}$$

The expectation values of these operators in a nuclear state $|\Psi\rangle$ are the electric and the magnetic moments of this state:

$$Q_{\lambda\mu} = \langle \Psi | \hat{Q}_{\lambda\mu} | \Psi \rangle, \tag{B.26}$$

$$M_{\lambda\mu} = \langle \Psi | \hat{M}_{\lambda\mu} | \Psi \rangle. \tag{B.27}$$

We discuss first the *selection rules* for $Q_{\lambda\mu}$ and $M_{\lambda\mu}$, because they do not depend on the special form of $|\Psi\rangle$. Since the strong and electromagnetic interactions conserve parity, it is possible to choose $|\Psi\rangle$ as an eigenstate of the parity operator. On the other side, the multipole operators $\hat{Q}_{\lambda\mu}$ and $\hat{M}_{\lambda\mu}$ have parity $(-)^{\lambda}$ and $(-)^{\lambda+1}$, respectively. This means that

$$\begin{aligned} Q_{\lambda\mu} &= 0 \quad \text{for } \lambda = 1, 3, 5, \ldots, \\ M_{\lambda\mu} &= 0 \quad \text{for } \lambda = 0, 2, 4, \ldots. \end{aligned} \tag{B.28}$$

Using the fact that the operators $\hat{Q}_{\lambda\mu}$ and $\hat{M}_{\lambda\mu}$ are spherical tensor operators in the sense of [Ed 57, Eq. (5.2.1.)], we can apply the Wigner–Eckart theorem [Ed 57, Eq. (5.4.1)] for eigenstates of angular momentum $|I, M\rangle$,

$$Q_{\lambda\mu} = \langle IM | \hat{Q}_{\lambda\mu} | IM \rangle = (-)^{I-M} \begin{pmatrix} I & \lambda & I \\ -M & \mu & M \end{pmatrix} \langle I \| \hat{Q}_{\lambda} \| I \rangle, \tag{B.29}$$

$$M_{\lambda\mu} = \langle IM | \hat{M}_{\lambda\mu} | IM \rangle = (-)^{I-M} \begin{pmatrix} I & \lambda & I \\ -M & \mu & M \end{pmatrix} \langle I \| \hat{M}_{\lambda\mu} \| I \rangle,$$

and get

$$\left.\begin{aligned} Q_{\lambda\mu} \\ M_{\lambda\mu} \end{aligned}\right\} \neq 0 \qquad \text{only for } \mu = 0 \text{ and } 0 \leqslant \lambda \leqslant 2I. \tag{B.30}$$

Since the dependence of $Q_{\lambda\mu}$ and $M_{\lambda\mu}$ on the magnetic quantum number is trivial [(B.29), (B.30)], we usually only give the values for $M = I$.

The *magnetic dipole moment* is defined by

$$\mu = \sqrt{\frac{4\pi}{3}} \langle II|\hat{M}_{10}|II\rangle \tag{B.31}$$

and the *electric quadrupole moment* is given by

$$Q = \sqrt{\frac{16\pi}{5}} \langle II|\hat{Q}_{20}|II\rangle. \tag{B.32}$$

These values are defined in the laboratory systems and are measurable by experiment. For the internal quadrupole moment, see Eq. (1.70).

B.3 The Multipole Expansion of the Radiation Field

In the case of electromagnetic radiation, it is convenient to work in the transverse or Coulomb gauge [Ja 62, Chap. 6.5] with the conditions

$$\Phi = 0; \qquad \nabla \cdot \mathbf{A} = 0. \tag{B.33}$$

The Hamiltonian of the free field is then given by

$$H_{\text{field}} = \frac{1}{8\pi} \int \left(\frac{1}{c^2} \dot{\mathbf{A}}^2 + (\nabla \times \mathbf{A})^2 \right) d^3r, \tag{B.34}$$

which yields the wave equation

$$\Delta \mathbf{A} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A} = 0. \tag{B.35}$$

To represent the most general vector potential **A**, which obeys (B.33) and (B.35), we look for a complete orthogonal set that fulfills this requirement. At this point, one usually makes a Fourier transformation in space and time and keeps only the components perpendicular to the wave vector **k** in order to fulfill (B.33). Because of the spherical symmetry, we only apply the *Fourier transformation in time*, that is, we assume for our basis functions a time-dependence $\exp(-ikct)$, and obtain the Helmholtz equation

$$(\Delta + k^2)\mathbf{A} = 0. \tag{B.36}$$

According to the spherical symmetry of our problem, we look for solutions that are eigenfunctions of the orbital angular momentum operators $\mathbf{L}^2$ and L_z. A complete set of solutions that fulfill this condition and are regular at the origin, are given by $j_L(kr) \cdot Y_{LM_L}(\theta, \varphi)$, where $j_L(kr)$ are spherical Bessel functions (see [AS 65]) and Y_{LM} are the spherical harmonics [Ed 57]. Besides the orbital angular momentum **L**, the photon has a spin **S** with $S = 1$ according to the vector character of the potential **A**. The eigenfunctions of $\mathbf{S}^2$ and S_z are the vectors $\mathbf{e}_{M_S}$ ($M_S = +1, 0, -1$) [Ed 57, Eq. (5.9.4)]. The total angular momentum of the photon is given by $\mathbf{I} = \mathbf{L} + \mathbf{S}$. With the rules of angular momentum coupling, we find as eigenfunctions of $\mathbf{L}^2, \mathbf{S}^2$

$\mathbf{I}^2, I_z$ [Ed 57, Eq. (5.9.10)] the *vector spherical harmonics*:

$$\mathbf{Y}_{ILm} = \sum_{M_S M_L} C^{L\ 1\ I}_{M_L M_S m} Y_{LM_L}(\theta, \varphi)\mathbf{e}_{M_S}. \tag{B.37}$$

They form a complete orthogonal set

$$\int_{4\pi} d\cos\theta\, d\varphi \mathbf{Y}^*_{ILm}(\theta, \varphi) \cdot \mathbf{Y}_{I'L'm'}(\theta, \varphi) = \delta_{II'}\delta_{LL'}\delta_{mm'}.$$

For fixed I, the orbital angular momentum can be $I-1$, I, $I+1$. That means there are three independent solutions of (B.36) for each k, I, m. However we have to fulfill the transversality condition (B.33).

From [Ed. 57, Eq. (5.9.14)],

$$\mathbf{Y}_{II'}(\theta, \varphi) = \frac{1}{\sqrt{I(I+1)}} \mathbf{L} Y_{Im}(\theta, \varphi) \tag{B.38}$$

we see that Eq. (B.33) is fulfilled for $j_I(kr)Y_{IIm}(\theta, \varphi)$. The orthogonal wave functions to (B.38) are linear combinations of $\mathbf{Y}_{II+1m}$ and $\mathbf{Y}_{II-1m}$. There is only one such combination that has the property (B.33) (see [Ed 57, Eq. (5.9.18)ff] and [AS 65, Eq. (10.1.21)ff]):

$$\frac{i}{k}\nabla \times j_I(kr)\mathbf{Y}_{IIm} = \sqrt{\frac{I}{2I+1}}\ j_{I+1}(kr)\mathbf{Y}_{II+1m} - \sqrt{\frac{I+1}{2I+1}}\ j_{I-1}(kr)\mathbf{Y}_{II-1m}. \tag{B.39}$$

We therefore have for each I and m two independent solutions of the wave equation (B.36) which fulfill the transversality condition (B.33). We characterize them by the index λ and call them electric ($\lambda = E$) and magnetic ($\lambda = M$) radiation.

$$\begin{aligned}\mathbf{A}_{MkIm}(\mathbf{r}) &= \mathfrak{N} j_I(kr)\mathbf{Y}_{IIm}(\theta, \varphi) \\ &= \frac{\mathfrak{N}}{\sqrt{I(I+1)}} \frac{1}{i}(\mathbf{r} \times \nabla) j_I(kr) Y_{Im}(\theta, \varphi)\end{aligned} \tag{B.40}$$

and

$$\begin{aligned}\mathbf{A}_{EkIm}(\mathbf{r}) &= \frac{i}{k}(\nabla \times \mathbf{A}_{MkIm}(\mathbf{r})) \\ &= \frac{-\mathfrak{N}}{\sqrt{I(I+1)}} \frac{1}{k} \left\{ \nabla \left(Y_{Im}(\theta, \varphi) \frac{\partial}{\partial r}(r j_I(kr)) \right) \right. \\ &\qquad \left. + k^2 \mathbf{r} j_I(kr) Y_{Im}(\theta, \varphi) \right\}.\end{aligned} \tag{B.41}$$

Equation (B.41) is obtained from (B.40) by evaluating $\nabla \times (\nabla \times \mathbf{r}) j_I Y_{Im}$ and using the fact that $j_I Y_{Im}$ satisfies Eq. (B.36). $\mathfrak{N}$ is a normalization constant. The quantum number k is discretized by requiring the proper boundary conditions at a perfectly conducting sphere with radius R. This sphere is

introduced only to define spherical boundary conditions, and at the end, R approaches large values compared to the nuclear radius R_0. According to the laws of classical electrodynamics, the boundary conditions at the surface are [Ja 62] $E_{\parallel}=0$ and $B_{\perp}=0$. These conditions are fulfilled only by a discrete number of k-values, k_n. In the magnetic case they are given as the zeros of the spherical Bessel function,

$$j_I(k_nR)=0. \tag{B.42}$$

In the electric case, we find the condition

$$\frac{\partial}{\partial r}(rj_I(k_nr))\Big|_{r=R}=0. \tag{B.43}$$

In the following we use k as a discrete index, which runs over the solutions k_n of Eq. (B.42) or (B.43), respectively.

The normalization constant $\mathfrak{N}$ is chosen in such a way that the following orthogonality relation holds [AS 65 Eq. (11.4.5)].

$$\int_0^R r^2\,dr\int_{4\pi}d\cos\theta\,d\varphi\,\mathbf{A}^*_{\lambda'k'I'm'}(\mathbf{r})\cdot\mathbf{A}_{\lambda kIm}(\mathbf{r})=\frac{2\pi\hbar c}{k}\delta_{\lambda\lambda'}\delta_{kk'}\delta_{II'}\delta_{mm'}. \tag{B.44}$$

In the limit of large R-values, $\mathfrak{N}$ turns out to be [AS 65]

$$\mathfrak{N}=\sqrt{\frac{4\pi\hbar ck}{R}} \tag{B.45}$$

for both λ-values.

The properties of magnetic and electric radiation under the parity transformation Π can be seen from (B.38). Since $\mathbf{L}$ preserves the parity and the curl operation changes parity, we find that

$$\Pi\mathbf{A}_{EkIm}=(-)^{I+1}\cdot\mathbf{A}_{EkIm};\qquad \Pi\mathbf{A}_{MkIm}=(-)^{I}\cdot\mathbf{A}_{MkIm}. \tag{B.46}$$

The most general solution of our problem is therefore given by

$$\mathbf{A}(\mathbf{r},t)=\sum_{\lambda k}\sum_{Im}\left\{a^*_{\lambda kIm}\mathbf{A}_{\lambda kIm}(\mathbf{r})e^{-ikct}+\text{c.c.}\right\}. \tag{B.47}$$

The coefficients $a_{\lambda kIm}$ are independent variables for the description of the electromagnetic field $\mathbf{A}$. The form (B.47) of $\mathbf{A}(\mathbf{r},t)$ guarantees that (B.33) and (B.36) are fulfilled for each choice of $a_{\lambda kIm}$.

Using Eq. (B.44), we can express the Hamiltonian (B.34) of the free field by

$$H_{\text{field}}=\sum_{\lambda k}\sum_{Im}(\hbar ck)\tfrac{1}{2}(a^*_{\lambda kIm}a_{\lambda kIm}+\text{c.c.}). \tag{B.48}$$

This is the Hamiltonian of harmonic oscillators $H_{ho}=\hbar\omega a^*a$, and the rules of *canonical quantization* can be applied: The variables a^*_λ and a_λ are replaced by creation and annihilation operators $\hat{a}^+_\lambda$ and $\hat{a}_\lambda$ which obey the boson commutation relations

$$\begin{aligned}&[\hat{a}_{\lambda kIm},\hat{a}_{\lambda'k'I'm'}]=0,\\ &[\hat{a}_{\lambda kIm},\hat{a}^+_{\lambda'k'I'm'}]=\delta_{\lambda\lambda'}\delta_{kk'}\delta_{II'}\delta_{mm'}.\end{aligned} \tag{B.49}$$

The operator $\hat{a}^+_{\lambda kIm}$ creates a photon of the type λ (magnetic or electric radiation) with the energy $\hbar ck$ and the angular momentum quantum numbers I, m. Applying the rules of second quantization (Appendix C), we find for the Hamiltonian

$$H_{\text{field}} = \sum_{k\lambda} \sum_{Im} (\hbar ck)\{\hat{a}^+_{\lambda kIm}\hat{a}_{\lambda kIm} + \tfrac{1}{2}\}. \tag{B.50}$$

The eigenstates can be represented in the occupation number representation $|\ldots n_{\lambda kIm}\ldots\rangle$, and the potential A (B.47) goes over into the operator

$$\hat{\mathbf{A}}(\mathbf{r}, t) = \sum_{\lambda kIm} \{\mathbf{A}_{\lambda kIm}(\mathbf{r})e^{-ickt}\hat{a}^+_{\lambda kIm} + \mathbf{A}^*_{\lambda kIm}(\mathbf{r})e^{ickt}\hat{a}_{\lambda kIm}\}. \tag{B.51}$$

B.4 Multipole Transitions

Together with the eigenstates $|\Psi_i\rangle$ of the nuclear Hamiltonian H_{nucl}, we now have a complete system of eigenstates of the unperturbed Hamiltonian

$$H_0 = H_{\text{nucl}} + H_{\text{field}}. \tag{B.52}$$

The time-dependent interaction (B.4)

$$H_{\text{int}} = -\frac{1}{c}\sum_{\lambda kIm}\left\{\hat{a}^+_{\lambda kIm}\int \mathbf{j}\cdot\mathbf{A}_{\lambda kIm}\, d^3\mathbf{r}\, e^{-ikct} + \text{h.c.}\right\} \tag{B.53}$$

does not commute with the photon number. It causes transitions between nuclear states $|\Psi_i\rangle$ and $|\Psi_f\rangle$ and the corresponding absorption and emission of photons

$$|i\rangle = |\Psi_i\rangle|\ldots n_\lambda\ldots\rangle \begin{array}{l} \xrightarrow{\text{absorption}} |\Psi_f\rangle|\ldots n_\lambda - 1\ldots\rangle = |f\rangle, \\ \xrightarrow[\text{emission}]{} |\Psi_f\rangle|\ldots n_\lambda + 1\ldots\rangle = |f\rangle. \end{array}$$

We calculate the transition probability in first-order Dirac perturbation theory [Me 61]:

$$T_{fi} = \frac{2\pi}{\hbar}|\langle f|H_{\text{int}}|i\rangle|^2 g(E_f), \tag{B.54}$$

where $g(E_f)$ is the number of final states per energy unit. We have to calculate T_{fi} at the energy $c\hbar k = E_f - E_i$ of the absorbed or emitted photon.

The relations (C.12)

$$|\langle\ldots n_\lambda - 1\ldots|\hat{a}_\lambda|\ldots n_\lambda\ldots\rangle|^2 = n_\lambda \tag{B.55}$$

and

$$|\langle\ldots n_\lambda + 1\ldots|\hat{a}^+_\lambda|\ldots n_\lambda\ldots\rangle|^2 = n_\lambda + 1 \tag{B.56}$$

show that the probability of an absorption of a photon is proportional to n_λ, and the probability of emission of a photon is proportional to $n_\lambda + 1$. Even for a vanishing field ($n_\lambda = 0$) we therefore have a probability for the

emission of a photon (spontaneous emission). In the following, we will only consider this case. It is identical to the absorption of a photon in a field with $n_\lambda = 1$.

The density of states $g(k)$ can be calculated from the condition (B.42) in the magnetic case (the condition (B.43) in the electric case will give the same result). In the limit of large R-values, we find

$$0 = j_I(k_n R) \simeq \frac{1}{k_n R} \sin\left(k_n R - \frac{\pi}{2} I\right) \tag{B.57}$$

that is,

$$k_n R = n \cdot \pi + \frac{\pi}{2} \cdot I, \qquad n = 0, \pm 1, \pm 2, \ldots, \tag{B.58}$$

and

$$g(k) = \frac{1}{\Delta E(k)} = \frac{1}{\hbar c} \frac{1}{\Delta k} = \frac{1}{\hbar c} \frac{R}{\pi}. \tag{B.59}$$

Using (B.54), (B.40), and (B.45), we get for the probability of the emission of a single photon with the quantum numbers (λ, k, I, m),

$$T_{fi}(E, kIm) = \frac{8\pi k}{\hbar} \left| \langle f | \frac{1}{ck} \int \mathbf{j} \cdot \nabla \times j_I(kr) \mathbf{Y}_{IIm} \, d^3 r | i \rangle \right|^2, \tag{B.60}$$

$$T_{fi}(M, kIm) = \frac{8\pi k}{\hbar} \left| \langle f | \frac{1}{c} \int \mathbf{j} \cdot j_I(kr) \mathbf{Y}_{IIm} \, d^3 r | i \rangle \right|^2, \tag{B.61}$$

which can be written as

$$T_{fi}(\lambda, kIm) = \frac{8\pi(I+1)}{\hbar \cdot I((2I+1)!!)^2} \left(\frac{E_\gamma}{\hbar c} \right)^{2I+1} |\langle f | \hat{\mathfrak{M}}(\lambda k I m) | i \rangle|^2, \tag{B.62}$$

where the general multipole transition operator $\hat{\mathfrak{M}}(\lambda, kIm)$ in the electric case is given by (B.60) and (B.41)

$$\hat{\mathfrak{M}}(E, kIm) = \frac{i(2I+1)!!}{ck^{I+1}(I+1)} \int \mathbf{j} \cdot \nabla \times (\nabla \times \mathbf{r})(j_I(kr) \, Y_{Im}(\theta, \varphi)) \, d^3 r. \tag{B.63}$$

Evaluating the $\nabla \times (\nabla \times \mathbf{r})$ operator as in (B.41), we get two terms. In the first part, we can use the continuity equation after integration by parts,

$$\nabla \mathbf{j} = -\frac{\partial}{\partial t} \rho = ikc\rho, \tag{B.64}$$

which results from (B.8). Finally, we obtain:

$$\hat{\mathfrak{M}}(E, kIm) = \frac{(2I+1)!!}{k^I(I+1)} \int \left\{ \rho Y_{Im} \frac{\partial}{\partial r} r j_I(kr) + i \frac{k}{c} \mathbf{j} \cdot \mathbf{r} \, Y_{Im} j_I(kr) \right\} d^3 r. \tag{B.65}$$

In the magnetic case, we have

$$\hat{\mathfrak{M}}(M, kIm) = \frac{-(2I+1)!!}{ck^I(I+1)} \int \mathbf{j} \cdot (\mathbf{r} \times \nabla) \cdot (j_I(kr) \cdot Y_{Im}) \, d^3 r. \tag{B.66}$$

In nuclear physics we usually work in the limit of *long wavelengths*, that is, the wavelength of the radiation is large compared to the nuclear radius ($k \cdot R_0 \ll 1$), or

$$E_\gamma \ll \frac{\hbar c}{r_0} A^{-1/3} = 197 \cdot A^{-1/3} \,[\mathrm{MeV}]. \tag{B.67}$$

In this case we can use the small-argument limit for the spherical Bessel function,

$$j_I(kr) \simeq \frac{(kr)^I}{(2I+1)!!}\left(1 - \frac{1}{2}\frac{(kr)^2}{2I+3} + \cdots\right) \tag{B.68}$$

and get to first order in kr

$$\hat{\mathfrak{M}}(E, kIm) = \int \rho r^I Y_{Im}\, d^3r + \frac{ik}{I+1}\int (\mathbf{r}\times\boldsymbol{\mu})\nabla r^I Y_{Im}\, d^3r. \tag{B.69}$$

The first part does not depend on k and corresponds to the static *electric multipole operators* $\hat{Q}_{Im}$ (B.18). The second term, of first order in kR_0, is usually neglected.

For the *magnetic operators* in this limit, we obtain exactly the operator (B.19) and (B.22)

$$\hat{\mathfrak{M}}(M, kIm) = \frac{1}{c(I+1)}\int (\mathbf{r}\times\mathbf{j})\cdot\nabla\left(r^I Y_{Im}\right) d^3r = \hat{M}_{Im}. \tag{B.70}$$

Usually we do not distinguish between the different orientations of the angular momenta (the exception being polarization measurements). We therefore assume a statistical distribution in the initial state and calculate the arithmetical average of the initial m-values. All the final m-values have to be summed up and we finally get the total probability for a certain multipole transition,

$$T_{fi}(\lambda, I) = \frac{1}{2I_i+1}\sum_{m_i, m_f, m} T_{fi}(\lambda, kIm). \tag{B.71}$$

Using the Wigner–Eckart theorem (B.29) and the orthogonality relations of the $3j$-symbols [Ed 57, Eq. (3.7.8)], we can carry out the sum over m_i, m_f and get for the transition probability (B.71)

$$T_{fi}(\lambda, I) = \frac{8\pi(I+1)}{\hbar I((2I+1)!!)^2}\left(\frac{E_\gamma}{\hbar c}\right)^{2I+1} B(\lambda I, I_i \to I_f). \tag{B.72}$$

The $B(\lambda)$ values are called *reduced transition probabilities* and are given by the reduced matrix elements

$$B(EI, I_i \to I_f) = \frac{1}{2I_i+1}|\langle f\|\hat{Q}_I\|i\rangle|^2, \tag{B.73}$$

$$B(MI, I_i \to I_f) = \frac{1}{2I_i+1}|\langle f\|\hat{M}_I\|i\rangle|^2, \tag{B.74}$$

where the multipole operators $\hat{Q}_{Im}$ and $\hat{M}_{Im}$ are defined in Eqs. (B.23) and (B.24). The $B(EI)$ and $B(MI)$ values contain the information about the

Table B.1 Transition probabilities T ($\sec^{-1}$) expressed by $B(EI)$ ($e^2(\mathrm{fm})^{2I}$) and $B(MI)$ ($\mu_N^2(\mathrm{fm})^{2I-2}$), and the Weisskopf units B_{sp} expressed in ($e^2(\mathrm{fm})^{2I}$) and ($\mu_N^2(\mathrm{fm})^{2I-2}$). The energies E are measured in MeV.

$T(\mathrm{E1}) = 1.587 \cdot 10^{15} \cdot \mathrm{E}^3 \cdot B(\mathrm{E1})$	$B_{sp}(\mathrm{E1}) = 6.446 \cdot 10^{-2} \cdot A^{2/3}$
$T(\mathrm{E2}) = 1.223 \cdot 10^{9} \cdot \mathrm{E}^5 \cdot B(\mathrm{E2})$	$B_{sp}(\mathrm{E2}) = 5.940 \cdot 10^{-2} \cdot A^{4/3}$
$T(\mathrm{E3}) = 5.698 \cdot 10^{2} \cdot \mathrm{E}^7 \cdot B(\mathrm{E3})$	$B_{sp}(\mathrm{E3}) = 5.940 \cdot 10^{-2} \cdot A^{2}$
$T(\mathrm{E4}) = 1.694 \cdot 10^{-4} \cdot E^9 \cdot B(\mathrm{E4})$	$B_{sp}(\mathrm{E4}) = 6.285 \cdot 10^{-2} \cdot A^{8/3}$
$T(\mathrm{E5}) = 3.451 \cdot 10^{-11} \cdot E^{11} \cdot B(\mathrm{E5})$	$B_{sp}(\mathrm{E5}) = 6.928 \cdot 10^{-2} \cdot A^{10/3}$
$T(\mathrm{M1}) = 1.779 \cdot 10^{13} \cdot E^3 \cdot B(\mathrm{M1})$	$B_{sp}(\mathrm{M1}) = 1.790$
$T(\mathrm{M2}) = 1.371 \cdot 10^{7} \cdot E^5 \cdot B(\mathrm{M2})$	$B_{sp}(\mathrm{M2}) = 1.650 \cdot A^{2/3}$
$T(\mathrm{M3}) = 6.387 \cdot 10^{0} \cdot E^7 \cdot B(\mathrm{M3})$	$B_{sp}(\mathrm{M3}) = 1.650 \cdot A^{4/3}$
$T(\mathrm{M4}) = 1.899 \cdot 10^{-6} \cdot E^9 \cdot B(\mathrm{M4})$	$B_{sp}(\mathrm{M4}) = 1.746 \cdot A^{2}$
$T(\mathrm{M5}) = 3.868 \cdot 10^{-13} \cdot E^{11} \cdot B(\mathrm{M5})$	$B_{sp}(\mathrm{M5}) = 1.924 \cdot A^{8/3}$

nuclear wave functions. They are often measured in Weisskopf units [see Eq. (B.85)]. The rest are kinematical factors. Table (B.1) gives the transition probabilities T_{fi} and the Weisskopf units for the $B(\lambda)$ values in the most important cases.

From the structure of the matrix elements

$$\langle I_f M_f | \hat{Q}_{Im} | I_i M_i \rangle \quad \text{and} \quad \langle I_f M_f | \hat{M}_{Im} | I_i M_i \rangle \tag{B.75}$$

we find the following *selection rules* for a transition from an initial state $|i\rangle = |I_i M_i\rangle$ to a final state $|f\rangle = |I_f M_f\rangle$:

$$|I_i - I_f| \leq I \leq I_i + I_f, \tag{B.76}$$

$$M_f - M_i = m, \tag{B.77}$$

$$\pi_i \pi_{\lambda Im} \pi_f = 1. \tag{B.78}$$

The parity of the multipole radiation (λI) $\pi_{\lambda Im}$ is given by the parity of the interaction $\propto \mathbf{A}_\lambda \cdot \mathbf{j}$. Since $\mathbf{j}$ changes sign under the parity transformation, we get from (B.46)

$$\pi_{EIm} = (-)^I, \qquad \pi_{MIm} = (-)^{I+1}. \tag{B.79}$$

From (B.72) and Table B.1 we find that the radiation with higher I-values is strongly suppressed. As long as selection rules or special effects in the nuclear structure part [$B(\lambda)$-values] do not forbid a transition, we usually have to take into account only the lowest possible I-value. Magnetic radiation is weaker than the electric radiation. Therefore, we often have a competition between M1 and E2 radiation.

From (B.72) it also becomes clear that the transition probability T increases rapidly with the transition energy. This is the reason why transitions with small energy differences are sometimes harder to observe. Other processes (like internal conversion) become important in such cases.

B.5 Single-Particle Matrix Elements in a Spherical Basis

In all microscopic calculations of electromagnetic moments and transition probabilities, we need the matrix elements of the multipole operators. We restrict ourselves to a spherical basis, that is, the single-particle states are given by the quantum numbers $|jm\rangle = |nsljm\rangle$. Since $s = \frac{1}{2}$, it is usually omitted, but the order in *slj* is important.

Using the phases of Edmonds [Ed 57], we obtain for the reduced single-particle matrix elements of the *electric multipole operators* (B.23) [RBS 73]

$$\langle f \| \hat{Q}_I \| i \rangle = \langle f \| e r^I Y_I(\theta, \varphi) \| i \rangle \tag{B.80}$$

$$= e \frac{\left(1 + (-)^{l_i + l_f + I}\right)}{2} \langle f | r^I | i \rangle \sqrt{\frac{(2I+1)(2j_i+1)(2j_f+1)}{4\pi}}$$

$$\times (-)^{j_f - \frac{1}{2}} \begin{pmatrix} j_f & I & j_i \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}. \tag{B.81}$$

In the *magnetic* case we have the spin part and the orbital part:

$$\langle f \| \hat{M}_I \| i \rangle = \langle f \| \mu_N \left(g_s \mathbf{s} + \frac{2}{I+1} g_l \mathbf{l} \right) \cdot (\nabla r^I Y_I) \| i \rangle$$

$$= \mu_N \frac{\left(1 - (-)^{l_i + l_f + I}\right)}{2} \langle f | r^{I-1} | i \rangle \sqrt{\frac{(2I+1)(2j_i+1)(2j_f+1)}{4\pi}}$$

$$\times (-)^{j_f - \frac{1}{2}} \begin{pmatrix} j_f & I & j_i \\ -\frac{1}{2} & 0 & +\frac{1}{2} \end{pmatrix} (I-k) \left[\tfrac{1}{2} g_s - g_l (1 + k/(I+1)) \right] \tag{B.82}$$

with

$$k = (j_i + \tfrac{1}{2})(-)^{l_i + j_i + \frac{1}{2}} + (j_f + \tfrac{1}{2})(-)^{l_f + j_f + \frac{1}{2}}.$$

The radial integrals are defined as

$$\langle f | r^I | i \rangle = \int_0^\infty R_f(r) r^I R_i(r) r^2 \, dr \tag{B.83}$$

and the g-factors are given in (B.24).

Weisskopf has used a rough estimate for the corresponding $B(\lambda)$-values [We 51]. In the radial integration (B.83), he replaces the wave function $R_f \simeq R_i$ in the nuclear interior ($r < R_0$) by a constant and gets

$$\langle r^I \rangle \simeq \frac{3}{I+3} R_0^I. \tag{B.84}$$

Furthermore, he uses $j_f = I + \frac{1}{2}$ and $j_i = \frac{1}{2}$. From Eqs. (B.81) and (B.73), in

the electric case we find

$$B(EI)\simeq\frac{e^2}{4\pi}\left(\frac{3}{I+3}\right)^2 R_0^{2I}=\frac{(1.2)^{2I}}{4\pi}\left(\frac{3}{I+3}\right)^2 A^{2I/3}\left[e^2(\mathrm{fm})^{2I}\right]. \quad \text{(B.85)}$$

In the magnetic case, Weisskopf estimates that the magnet $B(MI)$-value [We 51] is

$$B(MI)\simeq\frac{10}{\pi}\left(\frac{3}{I+3}\right)^2 R_0^{2I-2}\mu_N^2$$

$$=\frac{10}{\pi}(1.2)^{2I-2}\left(\frac{3}{I+3}\right)^2 A^{(2I-2)/3}\left[\mu_N^2\cdot(\mathrm{fm})^{2I-2}\right]. \quad \text{(B.86)}$$

Equations (B.85) and (B.86) give a very rough estimate for the single-particle transitions. They are often used as units (Weisskopf or single-particle units) for the actual $B(EI)$- and $B(MI)$-values.

B.6 Translational Invariance and Electromagnetic Transitions

Until now we have assumed that we have the exact wave functions of the nuclear system. Such wave functions are eigenfunctions of the linear momentum, and in all the electromagnetic transitions the linear momentum is conserved. For instance, the absorption of $E1$-radiation, which is given by the operator (B.23)

$$E1=\sqrt{\frac{3}{4\pi}}\ e\sum_{i=1}^{Z}\mathbf{r}_i=\sqrt{\frac{3}{4\pi}}\ e\sum_{i=1}^{A}\left(\tfrac{1}{2}-t_3^{(i)}\right)\mathbf{r}_i, \quad \text{(B.87)}$$

excites only protons. In the center of mass system the total linear momentum vanishes; that is, in that system the neutrons will have the opposite momentum. The effective elongation of the protons is therefore only $\mathbf{r}_i-\mathbf{R}$, where $\mathbf{R}$ is the center of mass. Instead of (B.87), we get

$$\sqrt{\frac{4\pi}{3}}\ E1=:e\mathbf{D}=e\sum_{i=1}^{Z}(\mathbf{r}_i-\mathbf{R})=-e\sum_{i=1}^{A}t_3^{(i)}(\mathbf{r}_i-\mathbf{R})=e\frac{NZ}{A}(\mathbf{R}_p-\mathbf{R}_n), \quad \text{(B.88)}$$

where $\mathbf{R}_p$ and $\mathbf{R}_n$ are the centers of mass for protons and neutrons, respectively.

As long as we have wave functions $|\Psi_i\rangle$ and $|\Psi_f\rangle$, which are eigenstates of the linear momentum, the replacement $\mathbf{r}=\mathbf{r}_i-\mathbf{R}$ has no influence on the matrix elements $\langle\Psi_f|E1|\Psi_i\rangle$, since $\langle\Psi_f|\mathbf{R}|\Psi_i\rangle$ vanishes in this case. In most applications, however, we use wave functions which violate translational symmetry. In this case $\langle\Psi_f|\mathbf{R}|\Psi_i\rangle$ gives spurious contributions and we have to use the operator $\mathbf{D}$ of Eq. (B.88) where this spurious contribution is subtracted on the average. It is equivalent to use the $E1$-operator with an effective charge of $(N/A)e$ for protons and $-(Z/A)e$ for neutrons. Corresponding effective charges have been calculated for transitions with

higher multipolarity [SF 74], and it was found that they give contributions of the order of $1/A$ to the normal transition operators, which can be neglected for heavy nuclei [see, for instance, Eq. (11.49)].

B.7 The Cross Section for the Absorption of Dipole Radiation

In Section B.4 we calculated the probability for the emission of a photon with quantum numbers (λklm). We are now interested in the absorption cross section for electric dipole radiation. We assume that the incoming photon is polarized in the z-direction. We use the occupation number representation of the electromagnetic field in plane waves [SF 74]:

$$\mathbf{A}(\mathbf{r}) = \frac{\hbar c}{2\pi} \sum_{\mu=1}^{2} \int d^3q \frac{\mathbf{e}_{\mathbf{q}\mu}}{\sqrt{\hbar\omega_q}} \left[e^{i\mathbf{q}\mathbf{r}} a_{\mathbf{q}\mu} + \text{h.c.} \right]. \tag{B.89}$$

$\mathbf{e}_{\mathbf{k}\mu}$ are the two polarization vectors of the photon. The initial wave function has the form

$$|i\rangle = a^+_{\mathbf{k}z}|\Psi_0\rangle. \tag{B.90}$$

Using Eq. (B.54), we obtain the excitation probability of the excited state with the energy E_f. Since there is now only one such final state, the density of final states $\rho_f(E)$ is a δ-function $\delta(E - E_f + E_0)$. The cross section is defined as the quotient of this excitation probability divided by the current density of incoming photons with this energy.

The photon operators $a_{\mathbf{q}\mu}, a^+_{\mathbf{q}'\mu'}$ obey the commutation relations

$$\left[a_{\mathbf{q}\mu}, a^+_{\mathbf{q}'\mu'} \right] = \delta(\mathbf{q} - \mathbf{q}')\delta_{\mu\mu'}, \tag{B.91}$$

that is, the corresponding plane waves are normalized as $(1/\sqrt{2\pi})^3 e^{i\mathbf{q}\mathbf{r}}$, that is, to a box of volume $(2\pi)^3$. The current density of incoming photons is therefore $c/(2\pi)^3$. For the cross section of an excitation of the final state we get

$$\begin{aligned}\sigma_f(E) &= \frac{2\pi}{\hbar} \frac{(2\pi)^3}{c} \left| \langle f| - \frac{1}{c} \int \mathbf{j} \cdot \mathbf{A}\, d^3r\, |i\rangle \right|^2 \delta(E - E_f + E_0) \\ &= \frac{(2\pi)^4}{\hbar c} \left(\frac{\hbar}{2\pi} \right)^2 \frac{1}{\hbar\omega_{\mathbf{k}}} \left| \langle \Psi_f | \int \mathbf{j} \cdot \mathbf{e}_{\mathbf{k}z} e^{i\mathbf{k}\mathbf{r}}\, d^3r\, |\Psi_0\rangle \right|^2 \cdot \delta(E - E_f + E_0). \end{aligned} \tag{B.92}$$

In the long wavelength limit we can replace $\mathbf{e}_{\mathbf{k}z} e^{i\mathbf{k}\mathbf{r}}$ by $\nabla(\mathbf{e}_{\mathbf{k}z} \cdot \mathbf{r} e^{i\mathbf{k}\mathbf{r}})$ and use the continuity equation, which gives, in analogy to (B.64),

$$\begin{aligned}\sigma_f(E) &= \frac{(2\pi)^2}{c^2} \frac{1}{k} \left| \langle \Psi_f | \frac{i}{\hbar} \int [H, \rho(\mathbf{r})] z e^{i\mathbf{k}\mathbf{r}}\, d^3r\, |\Psi_0\rangle \right|^2 \cdot \delta(E - E_f + E_0) \\ &= \frac{(2\pi)^2}{c^2} \frac{(E_f - E_0)^2}{\hbar^2 k} \left| \langle \Psi_f | e \sum_{i=1}^{Z} z_i e^{i\mathbf{k}\mathbf{r}_i} | \Psi_0 \rangle \right|^2 \cdot \delta(E - E_f + E_0). \end{aligned} \tag{B.93}$$

As in Section B.6, we have to replace $\mathbf{r}_i$ by $\mathbf{r}_i - \mathbf{R}$ for the dipole radiation, where only the first term in an expansion of $\exp(i\mathbf{k}(\mathbf{r}_i - \mathbf{R}))$ is kept, and get

$$\sigma_f(E) = \frac{4\pi^2 e^2}{\hbar c}(E_f - E_0)|\langle \Psi_f | D_z | \Psi_0 \rangle|^2 \delta(E - E_f + E_0). \qquad \text{(B.94)}$$

To get similar cross sections for higher multipolarity, we have to transform the representation (B.89) of the electromagnetic potential into the representation (B.51). This is shown in great detail in [SF 74].